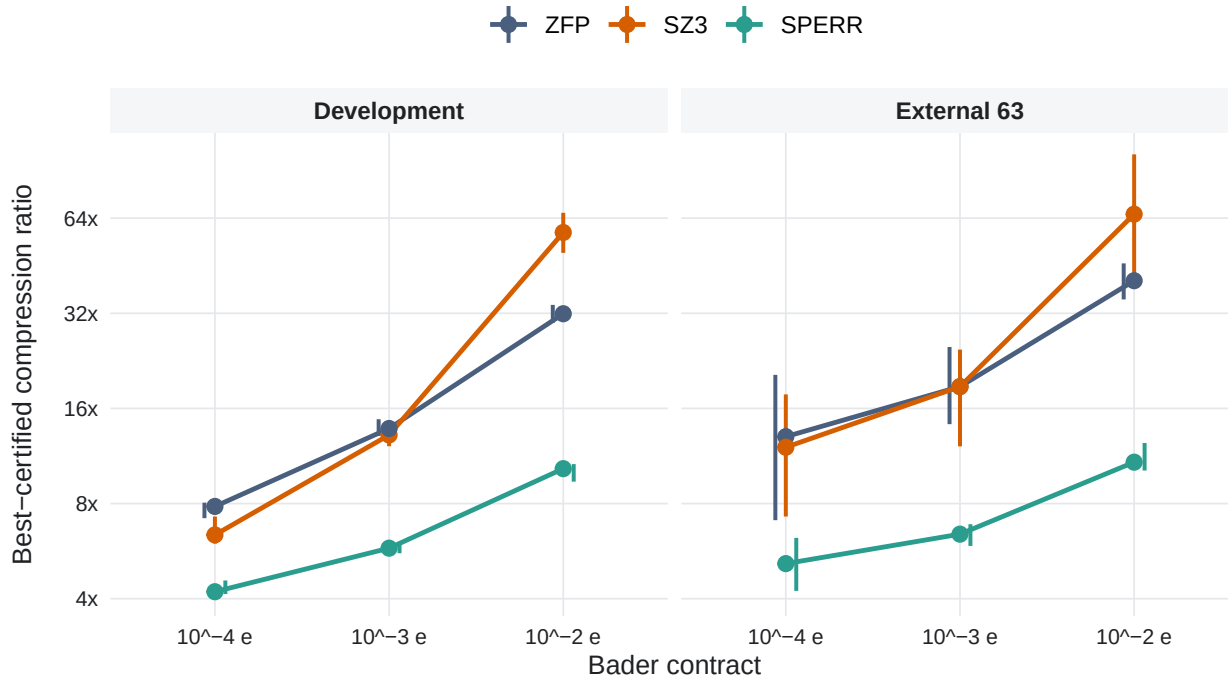


Supplementary Figure S8 | Frozen rate–fidelity conclusions reproduce on the untouched external cohort

Eligibility expands with tolerance and the codec ordering changes with the scientific contract; no protocol or scoring rule was retuned for external evaluation.

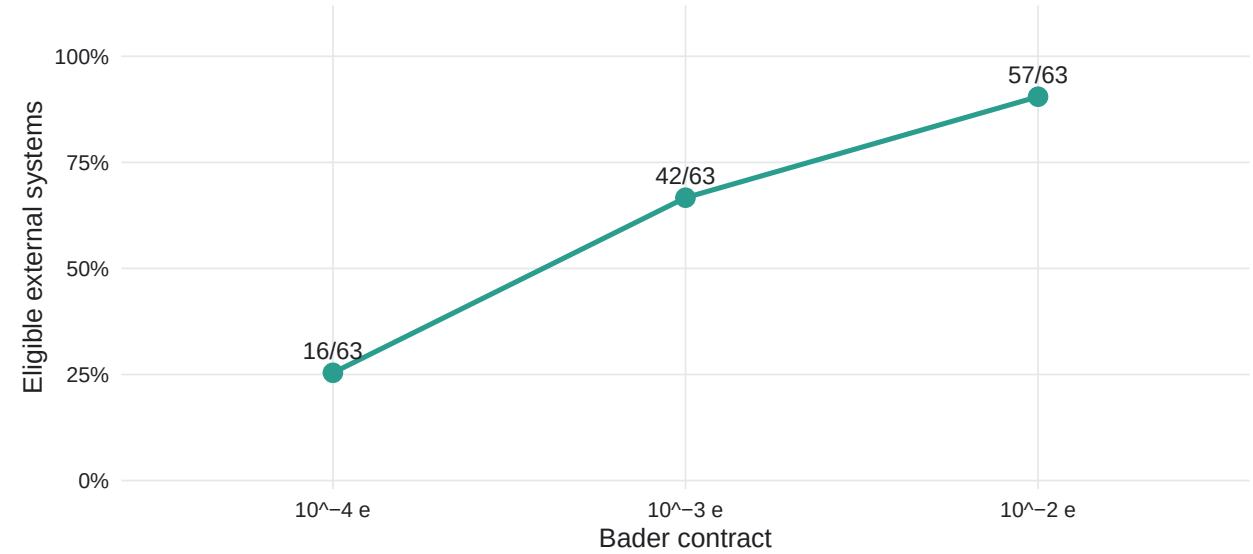
A | Rate–fidelity ordering changes with the scientific contract

Median best–certified compression ratio with 95% material–bootstrap CI



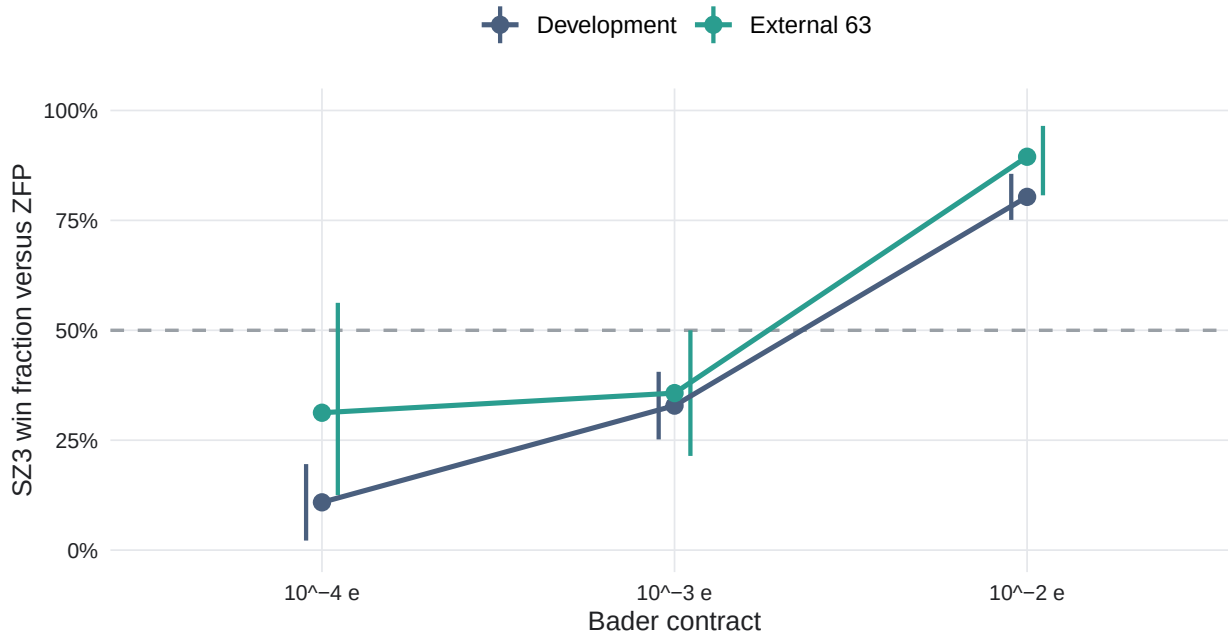
B | The externally measurable cohort expands as tolerance relaxes

Protocol A.1 eligibility; denominator is the untouched 63–system cohort



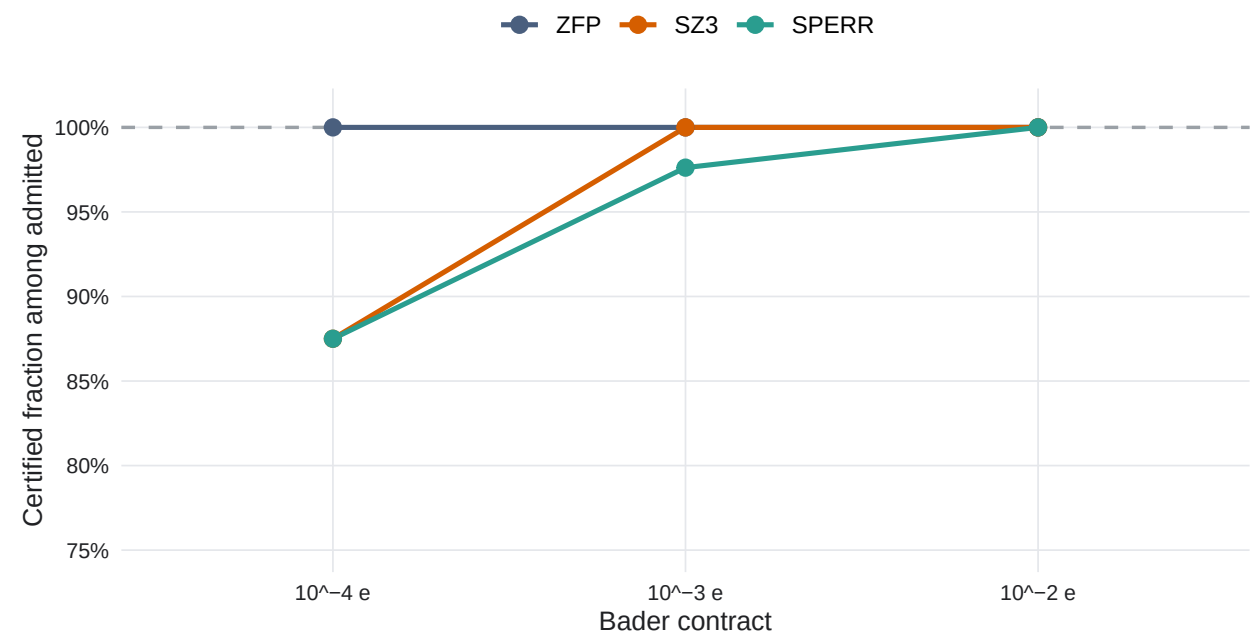
C | The SZ3–versus–ZFP crossover reproduces externally

Pairwise win fraction among eligible material comparisons; bars show bootstrap interval for SZ3 wins



D | Certification remains high after conditioning on eligibility

External certification fraction among Protocol A.1–admitted systems



A, pooled development and external rate–fidelity frontiers. B, external Protocol A.1 eligibility. C, pairwise SZ3–versus–ZFP crossover. D, certification conditional on eligibility.

The 63–system confirmatory cohort is distinct from the 65–system external descriptive/stability set; external rate–fidelity claims use only the frozen 63–system confirmatory outputs.